Max Destil

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OBJECTIVE

Seeking full-time employment opportunities to leverage my skills in PCB design, FPGA validation, embedded systems development and wireless communication systems gained through my graduate coursework and internship experiences.

EDUCATION

Rensselaer Polytechnic Institute, Troy, NY — Expected Graduation Date: December 2024
M.S. in Electrical Engineering, GPA: 3.65
Focus Area: Computer System Design
Relevant Coursework: Computer Architecture Prototyping with FPGAs, Microprocessor Systems, Advanced VLSI Design

Rensselaer Polytechnic Institute, Troy, NY — Graduated: December 2023
B.S. in Electrical Engineering, GPA: 3.53
Focus Area: Micro/Nano-electronics
Relevant Coursework: Microelectronics Technology, Optoelectronic Devices, Analog IC Design, Semi Power Electronics

EXPERIENCE

L3Harris Technologies, *Hardware Engineering Intern*, Grand Rapids, Michigan — May 2022 – August 2024

- Designed and validated multi-layer Printed Circuit Boards (PCBs) with Altium for safety-critical avionics systems.
- Led design of 8-Inch Multi-Functional Touchscreen Avionics Standby Device, delivering prototype in 10 weeks for AeroOne Summit where the product was presented to Airbus. Addressed signal integrity, thermal and power concerns.

Lighting Research Center, *Antenna Researcher*, Troy, New York — January 2022 – May 2023

- Conducted research on integrating 3D-printed antennas into wireless communication systems for lighting applications.
- Created Python scripts to resolve signal interference in 3D-printed antennas with BLE modules, improving detection accuracy by 40%, reducing latency to under 5 ms, extending range by 25%, and lowering power consumption by 20%.

PROJECTS

DC-DC Converter for Implantable Device Charging, *Graduate Project* — January 2024 – May 2024

- Designed and simulated a non-isolated DC-DC buck converter using PLECS achieving energy transfer efficiency of 92% through the implementation of synchronous rectification, and tuning switching frequencies to minimize losses.
- Achieved ±0.5% voltage regulation, 10mV ripple, and thermal output under 1W with transient / steady-state simulations.

FIR Filter with Pipelining & Parallel Processing, *Graduate Project* — January 2024 – May 2024

- Designed and implemented a 100-tap low-pass FIR filter in Verilog on FPGA, achieving 0.2π-0.23π rad/sample transition and 80dB attenuation.
- Optimized latency with pipelining and parallel processing (L=2, L=3), analyzing quantization effects for performance.

Real-Time Embedded System for Sensor Data Acquisition, *Undergraduate Project* — August 2023 – December 2023

- Developed real-time embedded systems in C on STM32 microcontrollers using FreeRTOS to manage multi-threaded tasks for sensor data acquisition.
- Integrated analog and digital signal processing with ADC/DAC modules improving signal processing accuracy by 20% and reducing system response time by 15% through optimized task scheduling and interrupt handling.

MindRace: EEG-Driven Robotic Car Control, *Passion Project* — June 2022 – June 2023

- Developed an embedded system integrating a NeuroSky EEG biosensor with Arduino and MSP432P401R microcontrollers to enable mind-controlled driving of Texas Instrument's TI-RSLK MAX robotic car.
- Processed EEG signals using Fast Fourier Transform to determine the car's speed based on the user’s focus levels.

SKILLS

PCB Design (Altium), FPGA (Verilog/SystemVerilog), Embedded Systems (C, FreeRTOS), Schematic/Layout Design, Signal/Power Integrity, I2C, SPI, UART, Lab Equipment (Oscilloscope), Python, C++, CAD tools (SolidWorks, KiCad)